Sitraka Randrianarivo

Clemson, SC — srandri@clemson.edu

EDUCATIONAL BACKGROUND

Expected May 2027 — **Ph.D. in Mathematical Sciences**
Clemson University. Advisor: Dr. Felice Manganiello

2023 — **M.S. in Mathematical Sciences**
Clemson University

2021 — **M.Sc. in Mathematical Sciences**
African Institute for Mathematical Sciences (AIMS) South Africa

2018 — **Bachelor's Degree in Pure Mathematics**
Faculty of Sciences, University of Antananarivo, Madagascar

PROFESSIONAL APPOINTMENTS

Clemson University, School of Mathematical and Statistical Sciences

2025–present — **Instructor of Record**
Math 1010: Essential Math for an Informed Society (Fall 2025)
Math 1040: Precalculus and Introduction to Differential Calculus (Spring 2026)
Math 1060: Calculus I (Fall 2026).

2022–present — **Graduate Teaching Assistant**
Linear Algebra (Math 3110); College Algebra (Math 1980);
Calculus I (Math 1060); Calculus of Several Variables (Math 2060);
Business Calculus I (Math 1020); Business Calculus II (Math 2070)

PUBLICATIONS

[1] Felice Manganiello and Sitraka Randrianarivo. Subspace code construction over the complex numbers using extended multiplicative characters. *Under review*, 2026. doi:10.13140/RG.2.2.33418.43208.

[2] Felice Manganiello and Sitraka Randrianarivo. Exact cardinality and nonredundant parametrization of character-polynomial codes. *Submitted. arXiv:2607.11595*, 2026.

[3] Sitraka Randrianarivo. Rational Reconstruction for Homological Construction. Master's thesis, African Institute for Mathematical Sciences, South Africa, 2021.

HONORS AND AWARDS

2021 — **Full Scholarship for M.S. Study**
African Institute for Mathematical Sciences (AIMS)South Africa.

TRAVEL SUPPORT

2026 — **Travel Funding for JMM**
Washington, DC

2023 — **Travel Funding for SIAM Southeastern Atlantic Section Annual Meeting**
Blacksburg, Virginia.

RESEARCH TALKS AND POSTERS

[1] Subspace codes from finite fields for the operator channel over complex number. In *2026 Joint Mathematics Meetings (JMM 2026)*. AMS, 2026.

[2] Subspace codes for Analogue Operator Channel, 2025. Special-session talk, 2025 Spring Southeastern Sectional Meeting, Clemson University, Clemson, SC.

[3] Construction of optimal codes for sum-rank weight, 2024. Poster presentation, Data sECurity and mAchine Learning (DECAL) Workshop, Clemson University, Clemson, SC.

RESEARCH MENTORSHIP

Graduate Research Mentor / TA, Clemson University

Summer 2022 Research Experience for Visiting Undergraduate (Math 4830).
Co-guided a group of visiting undergraduate researchers
through original investigations in sum-rank codes.

LANGUAGES

English: Fluent **French:** Fluent **Malagasy:** Native

OUTREACH AND COMMUNITY ENGAGEMENT

2026 **Volunteer**, Clemson Calculus Challenge

2024, 2026 **Math-In Volunteer**, Clemson University
Helped students prepare for final exams by answering questions on Calculus of Several Variables (Spring 2024), Precalculus, and Calculus I (Spring 2026).

TECHNICAL SKILLS

Mathematical typesetting	LaTeX
Computer algebra systems	SageMath, Macaulay2, Singular, Mathlab
Programming	Python, C